

BUOYANCIS

Next-Generation Commercial & Residential Rental Affordability Verification Protocol

Entity: BlackPalace AB

Location: Stockholm, Sweden

Target: Almi Loan & Growth Application

1. Executive Summary & Context

Buoyancis is an institutional-grade financial verification protocol designed to replace legacy, static rental affordability checks with a real-time, algorithmic present-value engine. By leveraging Open Banking APIs (PSD2 / ISO 20022 standard) and exponential time-decaying financial models, Buoyancis empowers property managers and landlords to evaluate applicant cash-flow health with unprecedented precision while protecting renters from rigid, outdated credit barriers.

Core Mission: To establish a standardized, tamper-proof, and inclusive affordability scoring infrastructure for European rental markets, mitigating credit default risk for property operators while expanding housing accessibility for modern dynamic earners.

2. Market Problem & Opportunity

The Nordic and European rental markets suffer from deep structural inefficiencies in tenant risk assessment:

- **Static Document Vulnerability:** Traditional vetting relies on past tax returns or paper payslips (3–6 months old). These static records fail to reflect sudden income shifts and are highly vulnerable to forgery or manual tampering.
- **Exclusion of Non-Traditional Earners:** Tech founders, freelancers, gig-economy professionals, and international recruits often lack traditional permanent employment contracts, leading to unjustified rejections despite strong net liquidity.
- **Fraud & Financial "Gaming":** Standard checks lack automated heuristics to identify short-term liquidity manipulation, such as round-trip internal transfers, artificial credit spikes, or temporary family loans intended to spoof bank balances.

3. Technical & Algorithmic Innovation

Buoyancis introduces a dynamic Present Value (PV) verification framework combined with deterministic anti-gaming heuristics:

A. Exponential Time-Decay Engine

Rather than averaging gross historical inflows, the engine applies a continuous time-discounting function $\$W(t)\$$, assigning exponentially higher weight to recent, settled inflows over older transactions:

$$PV = \sum [\text{Inflow}_i \times e^{(-\lambda \times \Delta t)}] \quad | \quad \lambda = 0.0077 \text{ (Half-Life } \approx 90 \text{ Days)}$$

B. Multi-Layer Anti-Gaming Heuristics

Rule Component	Mechanism	Risk Mitigated
Rule A: 90-Day Hard Gate	Evaluates cash flow within a strict moving 90-day window ($\Delta t \leq 90$).	Outdated financial claims & stale liquidity proofs.
Rule B: Outlier Cap	Caps single-transaction inflows at $3.0 \times \text{Average Inflow}$.	One-off artificial balance pumps & sudden capital injections.
Rule C: Eligibility Filter	Automated exclusion of internal self-transfers, refunds, and loan circularity.	Round-trip account manipulation and artificial volume inflating.

4. Market Strategy & Deployment Roadmap

Buoyancis is engineered as a B2B SaaS protocol integrated directly into leasing management workflows:

- Phase 1: Local Pilot Validation (Stockholm):** Partnering with commercial and residential property managers in Stockholm to conduct pilot verification runs, benchmarking accuracy against traditional manual checks.
- Phase 2: PropTech Platform Integration:** Exposing RESTful API endpoints and Next.js widget integrations for digital leasing platforms across the Nordics.
- Phase 3: Institutional Scale & Compliance:** Extending Open Banking integrations across Sweden and the wider EU under strict GDPR and financial data security protocols.

5. Sustainability & Value Creation

- **Social Impact:** Enables equitable access to housing for high-mobility, dynamic, and non-traditional earners, removing structural bias in tenant screening.
- **Economic Efficiency:** Drastically cuts default rates for landlords, reduces administrative overhead by over 80%, and eliminates manual document auditing.
- **Environmental Sustainability:** Fully digital, paperless protocol aligned with modern green PropTech standards.